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OPAMP_FULL & FDOPAMP_FULL

Parametrized op-amp and fully differential amplifier macromodels — examples of stability, noise and large-signal analysis

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THE MODELS

One parameter set, two topologies

OPAMP_FULL inp inn vcc vee out

Defaults: µA741-like, ±15 V, 1 MHz, 0.5 V/µs

FDOPAMP_FULL inp inn vcc vee outp outn vocm

Defaults: generic ±5 V, 100 MHz FDA with an internal CM loop

Linear

- A0, GBW and two extra poles
- input R and C, differential and CM
- CMRR(f), PSRR+(f), PSRR−(f)
- ROUT, COUT
- FDA: CM loop gain, bandwidth, pole

Noise

- white + 1/f input voltage noise
- white + 1/f current noise, per pin
- all internal resistors noiseless
- simulated = specified (≤ 0.1 %)

Nonlinear

- VOS, IB, IOS; optional input clamp
- tanh input stage: slew rates + smooth THD
- follower jump (input tail capacitance)
- input CM range, output swing limits
- current limit, class-AB crossover
- supply current follows the load

What the examples show that simple models miss

load-dependent output resistance (crossover) · an FDA common-mode loop that can be opened · noise exactly as specified · distortion that grows into the slew limit · swing limits that follow Vocm · the follower jump of the µA741

GETTING STARTED

Include the library, instantiate, override parameters

```
.include ../models/opamp_full.lib
X1 inp inn vcc vee out OPAMP_FULL
+ PARAMS: GBW=10Meg SR=5Meg ROUT=50

.include ../models/fdopamp_full.lib
X2 inp inn vcc vee outp outn vocm
+ FDOPAMP_FULL PARAMS: A0CM=30k
```

Every parameter has a default; pass only what differs. Units: SR in V/s, CMRR and PSRR as ratios (31.6k = 90 dB), noise in V/√Hz and A/√Hz.

Example (examples/)	Shows
01_opamp_cload_stability	loop gain, CL, crossover
02_fda_dm_cm_stability	DM and CM loops
03_opamp_noise	noise, NF vs Rs
04_opamp_large_signal	slew, THD, clipping
05_fda_swing_vocm	FDA swing vs Vocm
06_opamp_vom_vs_freq	max. output vs f
07_opamp_follower_pulse	follower jump (µA741)
08_opamp_follower_npn_pnp	jump: npn vs pnp

```
cd examples
python3 run_examples.py
```

models/
opamp_full.lib, fdopamp_full.lib

tests/
characterization decks

examples/
eight netlists and a runner

docs/
these slides, figures

EXAMPLE 1 · STABILITY

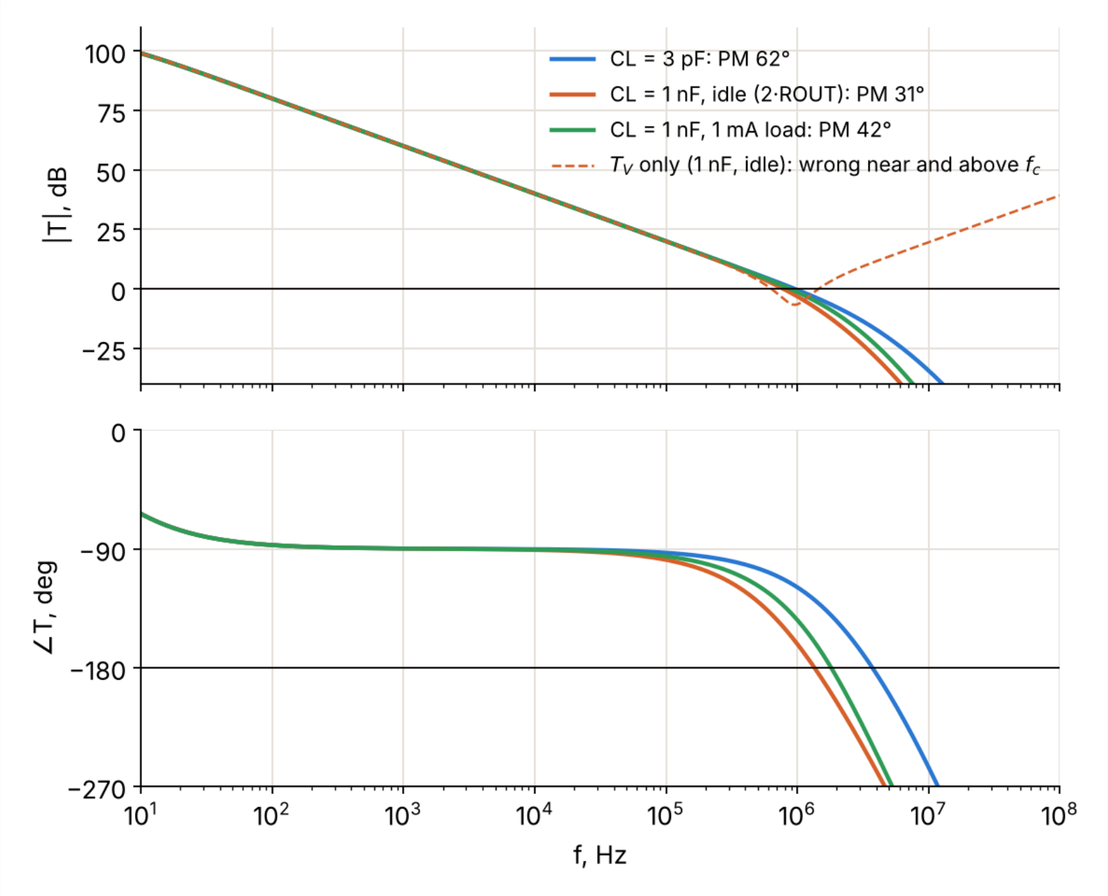
Follower with a capacitive load

```
X1 in f vcc vee o OPAMP_FULL
VI o f DC 0 AC 1 ; voltage injection
II 0 f DC 0 AC 0 ; current injection
CL f 0 1n
```

$$T = \frac{T_V\,T_I - 1}{T_V + T_I + 2}, \quad T_V = -\frac{V_o}{V_f}, \quad T_I = -\frac{I_o}{I_f}$$

Load	f(|T| = 1)	PM
CL = 3 pF	949 kHz	62°
CL = 1 nF, idle	777 kHz	31°
CL = 1 nF, 1 mA DC load	878 kHz	42°

Class-AB crossover: at idle the small-signal ROUT doubles, so 1 mA of DC load current adds 11° of phase margin.



EXAMPLE 2 · STABILITY

FDA: the common-mode loop is a second loop

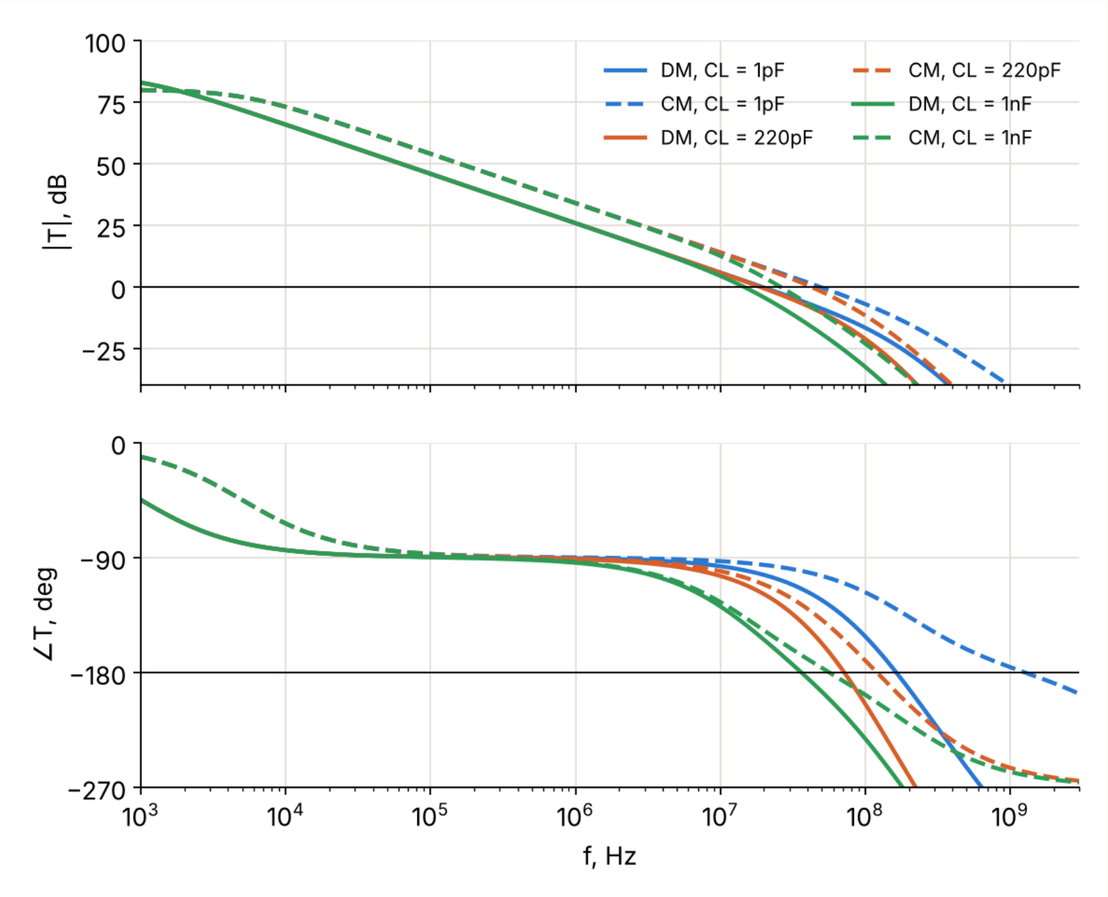
The CM loop is internal to the part. FDOPAMP_CORE brings its control path out as pins cmo → cmi, so it can be opened; FDOPAMP_FULL joins them.

```
XCMV np nn vcc vee op on vo co ci
+ FDOPAMP_CORE
Vh co ci DC 0 AC 1 ; inject at the hook
```

$$\frac{1}{1+T} = \frac{1}{1+T_V} + 1 + i$$

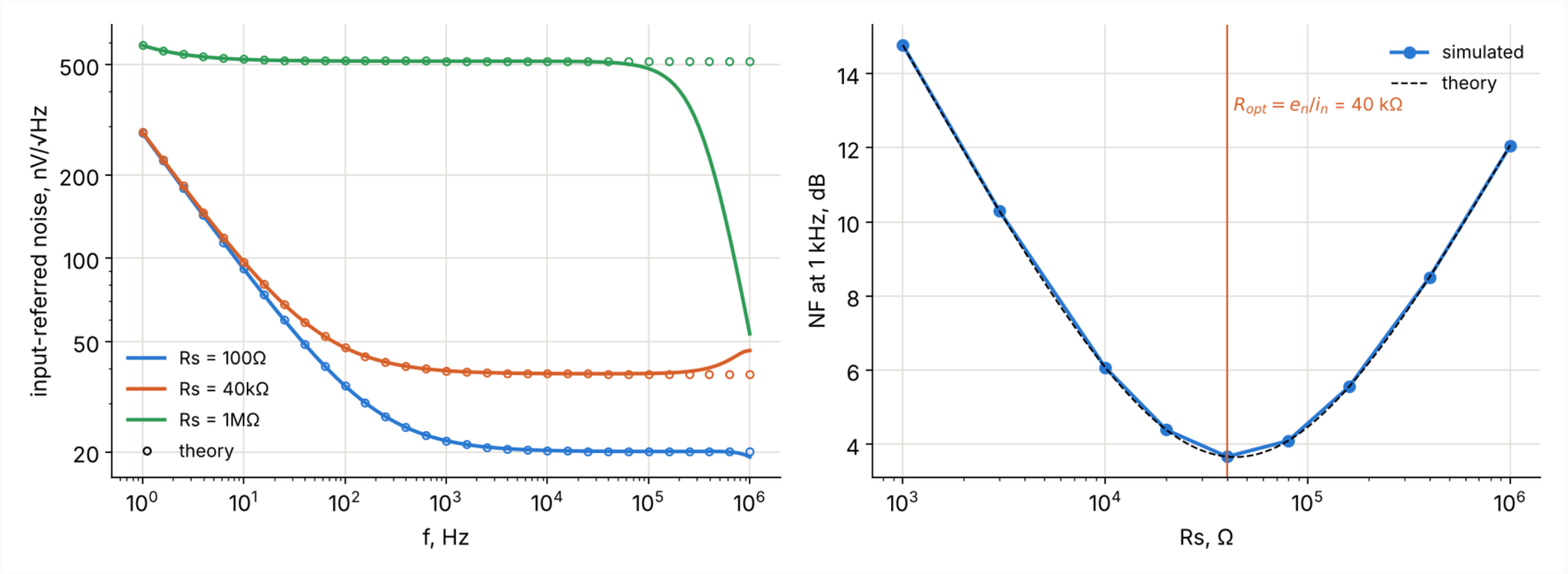
CL per output	DM: fc, PM	CM: fc, PM
1 pF	19.3 MHz, 77°	48.4 MHz, 76°
220 pF	18.7 MHz, 63°	42.1 MHz, 48°
1 nF	14.5 MHz, 39°	25.9 MHz, 24°

Load capacitance to ground hits the faster CM loop first.



EXAMPLE 3 · NOISE

Only the noise you specify



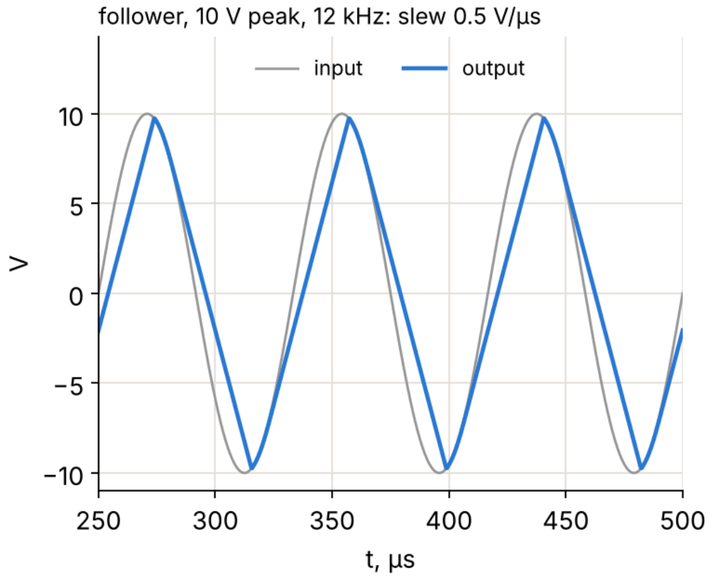
$$e_{\text{tot}}^2 = e_n^2\left(1 + \frac{f_{ce}}{f}\right) + i_n^2 R_s^2\left(1 + \frac{f_{ci}}{f}\right) + 4kTR_s \qquad R_{\text{opt}} = e_n/i_n = 40\ \text{k}\Omega$$

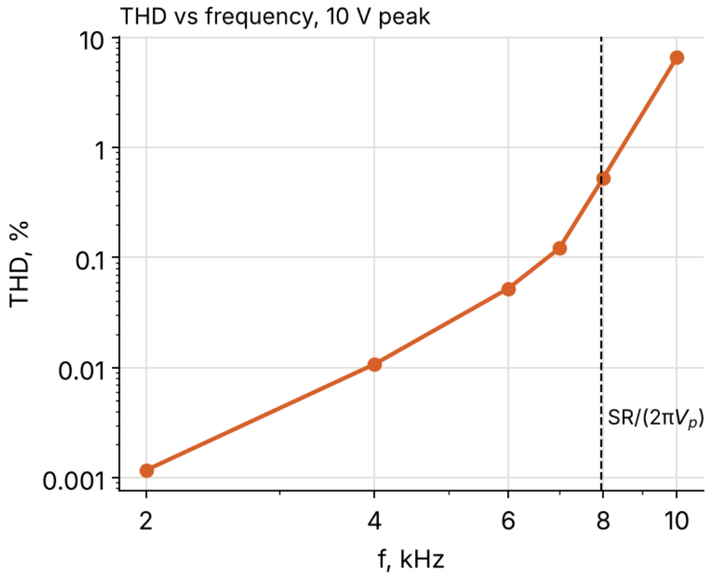
Defaults ENOISE = 20 nV/√Hz, FCEN = 200 Hz, INOISE = 0.5 pA/√Hz, FCIN = 0 (no 1/f current noise). Simulated / theory = 1.0000 from 1 Hz to 10 kHz for sources up to 40 kΩ; minimum noise figure 3.67 dB at 40 kΩ.

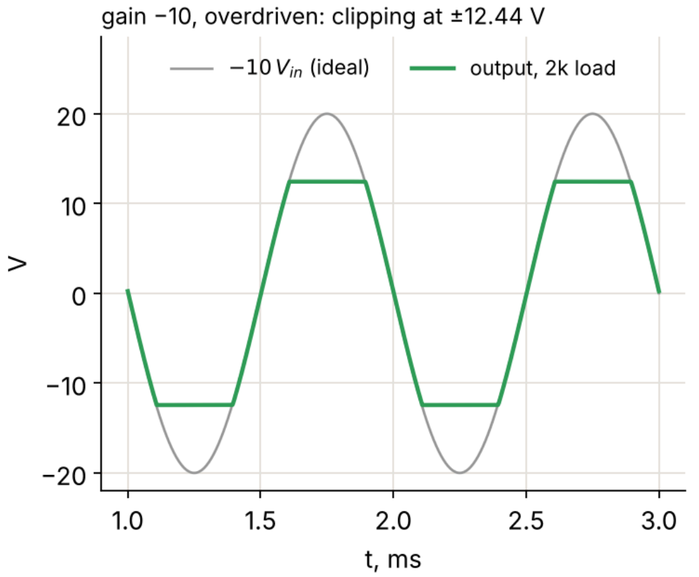
Above 100 kHz the departure is physical: the input capacitance (CIC = 0.75 pF per pin) low-passes the source noise (1 MΩ: 210 kHz), the closed loop rolls off near 1 MHz, and CID is no longer bootstrapped as the loop gain falls (+20 % at 40 kΩ).

EXAMPLE 4 · NONLINEAR EFFECTS

Slew rate, distortion and clipping from one model







Slew rate

The tanh input stage limits dV/dt to SR = 0.50 V/µs; no separate limiter.

Distortion

THD grows smoothly: 0.001 % at 2 kHz, 0.5 % at the full-power bandwidth

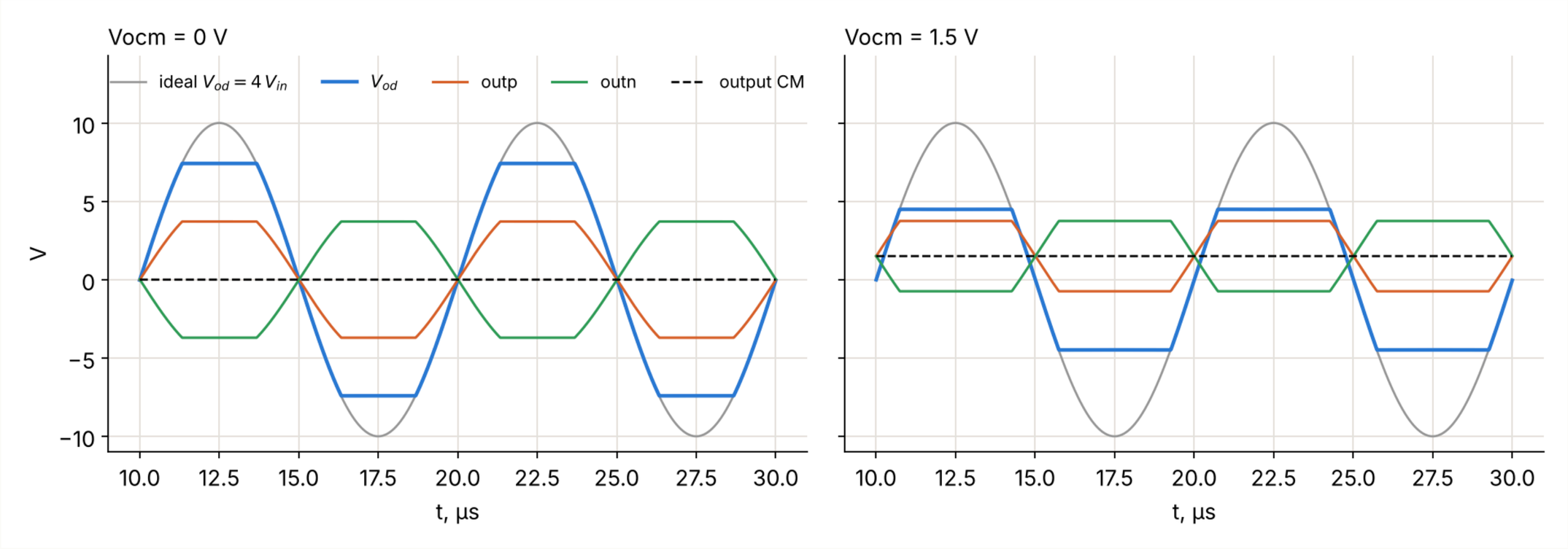
$$f_p = \frac{\mathrm{SR}}{2\pi V_p} = 8\ \mathrm{kHz}$$

Swing and current

Clips at ±12.44 V into 2 kΩ and recovers at once. 100 Ω load: 25 mA limit; supply current 26.7 mA.

EXAMPLE 5 · NONLINEAR EFFECTS

FDA output swing depends on Vocm

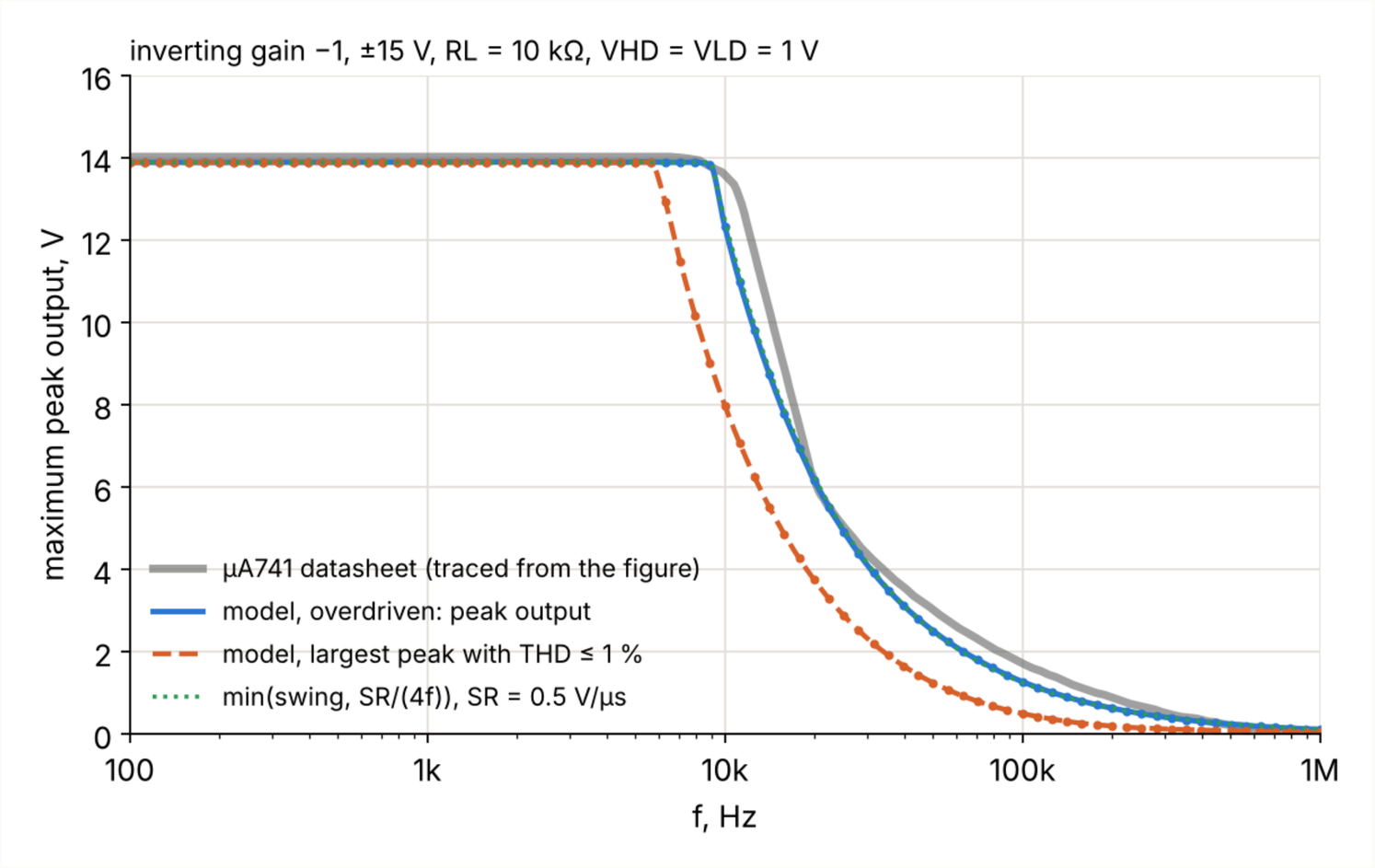


$$|V_{od}| \le 2\min\left(V_{cc} - V_{HD} - V_{ocm},\ V_{ocm} - V_{ee} - V_{LD}\right)$$

Vocm = 0: ±7.41 V; Vocm = 1.5 V: ±4.49 V. Output CM stays at Vocm within 1 mV while the DM path clips.

EXAMPLE 6 · NONLINEAR EFFECTS

Maximum output vs frequency: the µA741 datasheet curve



Two parameters set the curve

VHD/VLD give the flat ±13.9 V part; above about 8 kHz the output slews into a triangle of peak SR/(4f).

Against the datasheet

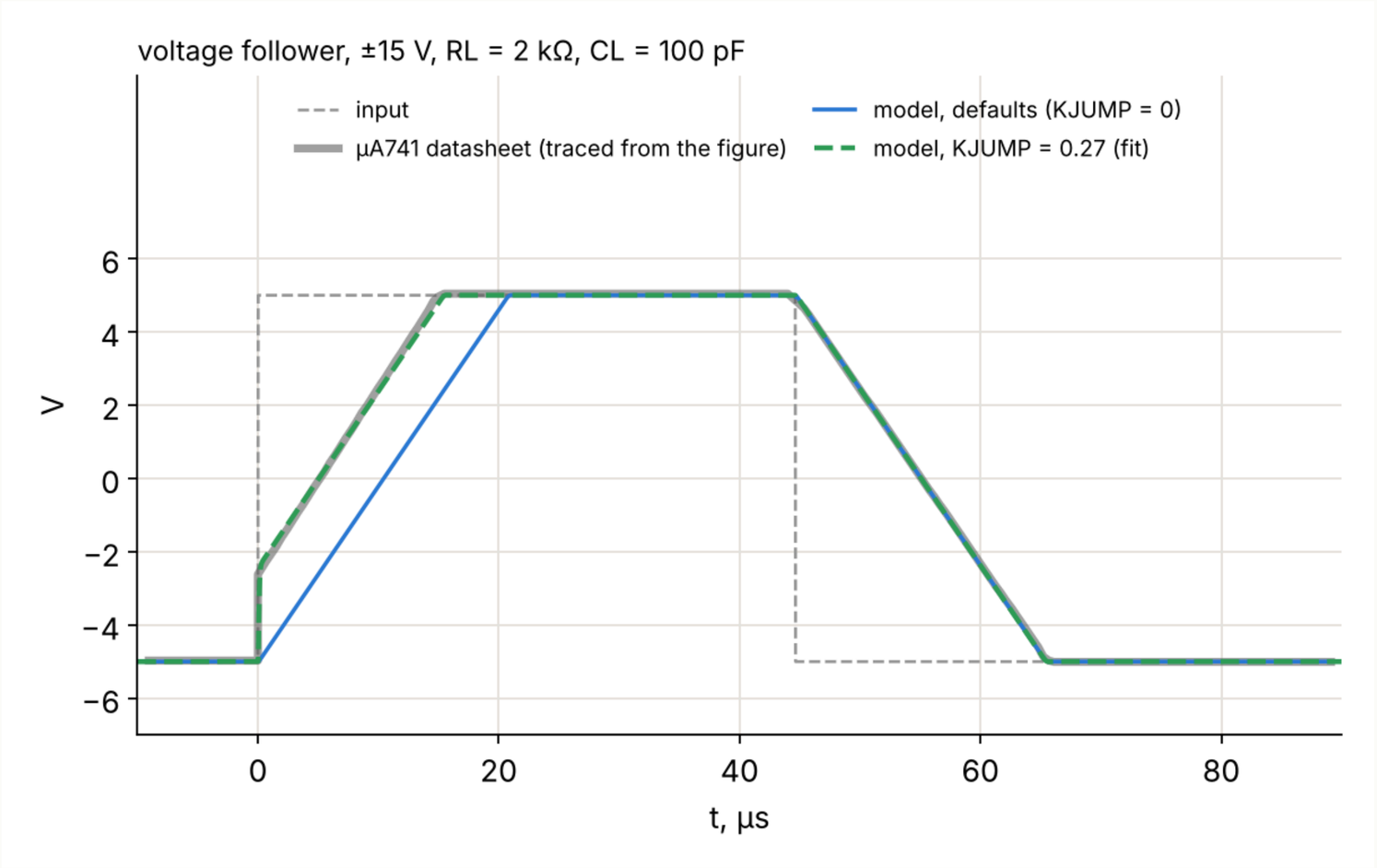
6.16 V at 20 kHz (datasheet 6.17 V). Above 50 kHz the datasheet is 20–40 % higher.

Distortion comes first

The largest peak with THD ≤ 1 % rolls off an octave earlier: the tanh stage bends before it slews.

EXAMPLE 7 · NONLINEAR EFFECTS

Follower pulse response: the jump and the slew rates



Tail-capacitance jump

A fast step drives the input pair's tail node; the output jumps by KJUMP·ΔV, then slews (Solomon 1974). A negative KJUMP moves it to the falling edge.

µA741 fit

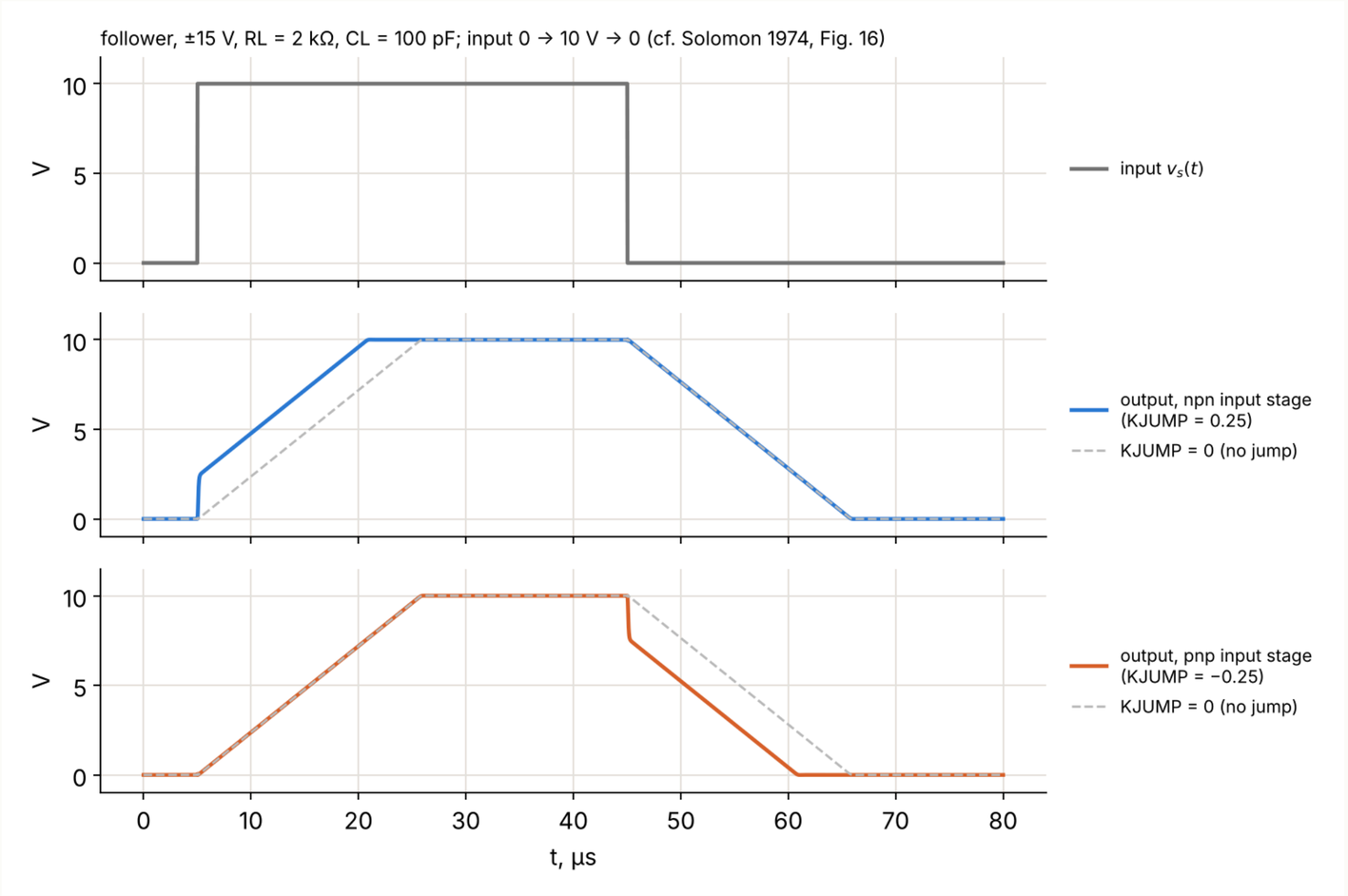
KJUMP = 0.27 with the default SR: RMS deviation 0.06 V (1.16 V without the jump).

Rising ≠ falling

SRR and SRF set the two slew rates, e.g. LF155 ≈ 5.5 / 11 V/µs; the small-signal gain is unchanged.

EXAMPLE 8 · NONLINEAR EFFECTS

The input-pair polarity picks the edge that jumps



npn: KJUMP = 0.25

The tail follows the higher input: the output jumps 2.38 V on the rising edge.

pnp: KJUMP = −0.25

The tail follows the lower input: the output drops 2.37 V on the falling edge.

As in Solomon (1974)

Theory: 0.964 · KJUMP · 10 V = 2.41 V. The other edge slews at 0.48 V/µs.

VERIFICATION

Characterization decks in tests/

OPAMP_FULL (defaults)

Quantity	Result
A0, fu, PM	106.0 dB, 956 kHz, 71°
CMRR, PSRR± (DC)	90 dB
Slew rate (10–90 %)	0.48 V/µs
Swing into 2 kΩ	±12.44 V
Current limit	±25.0 mA
Supply current	1.71 mA
Noise vs theory	< 0.02 %

FDOPAMP_FULL (defaults)

Quantity	Result
A0, fu, PM	100.0 dB, 95 MHz, 67°
CMRR (DC, 1 MHz)	90 dB, 70 dB
Slew DM / CM (10–90 %)	151 / 83 V/µs
Swing into 1 kΩ	±7.41 V diff.
Current limit	±49.7 mA
DM / CM loop, gain 4	77° / 76°
Noise vs theory	< 0.1 %

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